PVsyst V8.1.6

PVsyst - Simulation report

Grid-Connected System

Project: 52.2 KW solar PV plant for commercial building

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 52.2 kWp

Venkatesh Dream Hub, Dhule. - India

Author

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Project summary

Geographical Site
Venkatesh Dream Hub, Dhule.
India

Situation

Latitude	20.9042 °(N)
Longitude	74.7749 °(E)
Altitude	250 m
Time zone	UTC+5.5

Project settings
Albedo 0.20

Weather data
Venkatesh Dream Hub, Dhule.
Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1
Fixed plane
Tilt/Azimuth 12 / 0 °

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information

PV Array		Inverters	
Nb. of modules	90 units	Nb. of units	1 unit
Pnom total	52.2 kWp	Total power	50 kWac
		Pnom ratio	1.04

Results summary

Simulation step Hourly

Produced Energy	83078 kWh/year	Specific production	1592 kWh/kWp/year	Perf. Ratio PR	81.68 %

Table of contents

Project and results summary 2
General parameters, PV Array Characteristics, System losses 3
Main results 5
Loss diagram 6
Predef. graphs 7
P50 - P90 evaluation 8
Cost of the system 9
CO_2 Emission Balance 10

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

Simulation step
Hourly

Orientation #1
Fixed plane
Tilt/Azimuth 12 / 0 °

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module
Manufacturer Generic
Model ASB-M10-144-580-Bifacial-Gen2
(Original PVsyst database)
Unit Nom. Power 580 Wp

Inverter
Manufacturer Generic
Model SG50CX-P2
(Original PVsyst database)
Unit Nom. Power 50.0 kWac

Array #1 - PV Array

Number of PV modules	45 units	Number of inverters	2 * MPPT 25% 0.5 unit
Nominal (STC)	26.10 kWp	Total power	25.0 kWac
Modules	3 units x 15 In series		
At operating cond. (50°C)		Operating voltage	160-1000 V
Pmpp	24.01 kWp	Max. power (=>40°C)	55.0 kWac
U mpp	603 V	Pnom ratio (DC:AC)	1.04
I mpp	40 A		

Array #2 - Sub-array #2

Number of PV modules	45 units	Number of inverters	2 * MPPT 25% 0.5 unit
Nominal (STC)	26.10 kWp	Total power	25.0 kWac
Modules	3 units x 15 In series		
At operating cond. (50°C)		Operating voltage	160-1000 V
Pmpp	24.01 kWp	Max. power (=>40°C)	55.0 kWac
U mpp	603 V	Pnom ratio (DC:AC)	1.04
I mpp	40 A		

Total PV power

Nominal (STC)	52 kWp
Total	90 modules
Module area	232 m²
Cell area	215 m²

Total inverter power

Total power	50 kWac
Number of inverters	1 unit
Pnom ratio	1.04
No power sharing	

Array losses

Array Soiling Losses
Average loss Fraction 3.0 %

Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.
3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 29.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 247 mΩ
Global wiring resistance 124 mΩ
Loss Fraction 1.50 % at STC

LID - Light Induced Degradation
Loss Fraction 2.00 %

Array losses

Module Quality Loss
Loss Fraction -0.38 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

AC wiring losses

Inv. output line up to injection point
Inverter voltage 400 Vac tri
Loss Fraction 0.75 % at STC
Inverter: SG50CX-P2
Wire section (1 Inv.) Copper 1 x 3 x 16 mm²
Wires length 20 m

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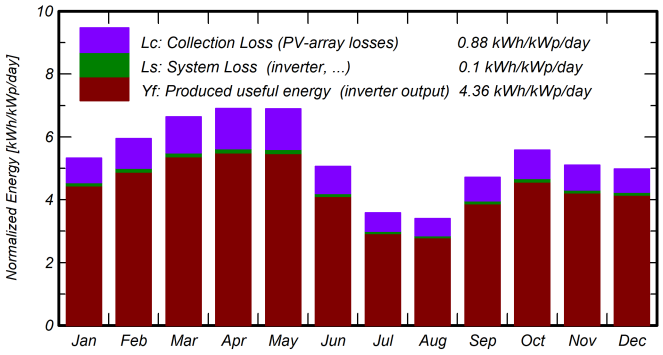
Main results

Simulation step
Hourly

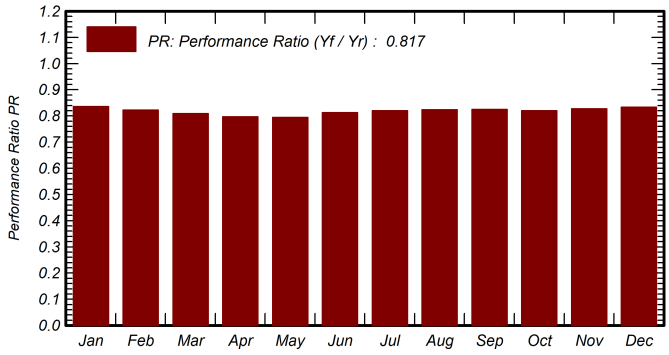
System Production

Produced Energy (P50)	83078 kWh/year	Specific production (P50)	1592 kWh/kWp/year	Perf. Ratio PR	81.68 %
Produced Energy (P90)	79897 kWh/year	Specific production (P90)	1531 kWh/kWp/year		
Produced Energy (P95)	79002 kWh/year	Specific production (P95)	1513 kWh/kWp/year		

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	141.0	43.72	21.00	165.1	156.4	7368	7201	0.836
February	148.6	47.23	24.10	166.4	158.0	7317	7149	0.823
March	193.3	65.33	28.24	205.9	195.8	8906	8696	0.809
April	204.3	73.75	32.37	207.1	197.0	8826	8617	0.797
May	219.7	80.17	33.64	213.8	202.9	9084	8869	0.795
June	157.7	92.96	30.64	151.8	143.3	6587	6442	0.813
July	114.5	75.61	27.89	111.0	104.6	4863	4754	0.820
August	106.5	68.26	26.84	105.4	99.4	4636	4529	0.823
September	137.4	77.91	27.11	141.3	133.8	6219	6080	0.824
October	158.8	65.86	26.86	173.0	164.0	7577	7406	0.820
November	133.8	52.18	24.38	153.1	145.1	6766	6614	0.827
December	130.8	44.64	21.66	154.5	146.3	6876	6721	0.833
Year	1846.2	787.62	27.07	1948.6	1846.9	85025	83078	0.817

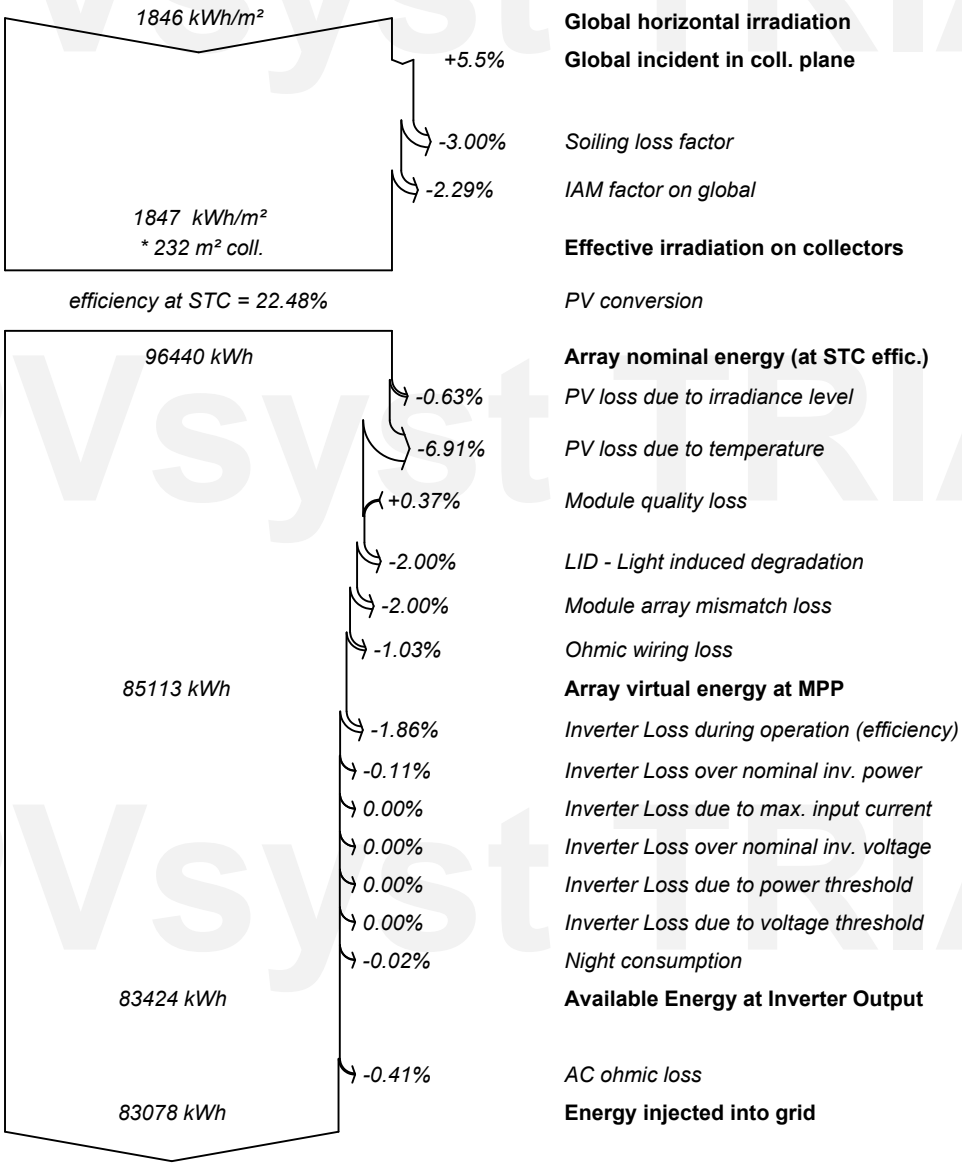
Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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Loss diagram



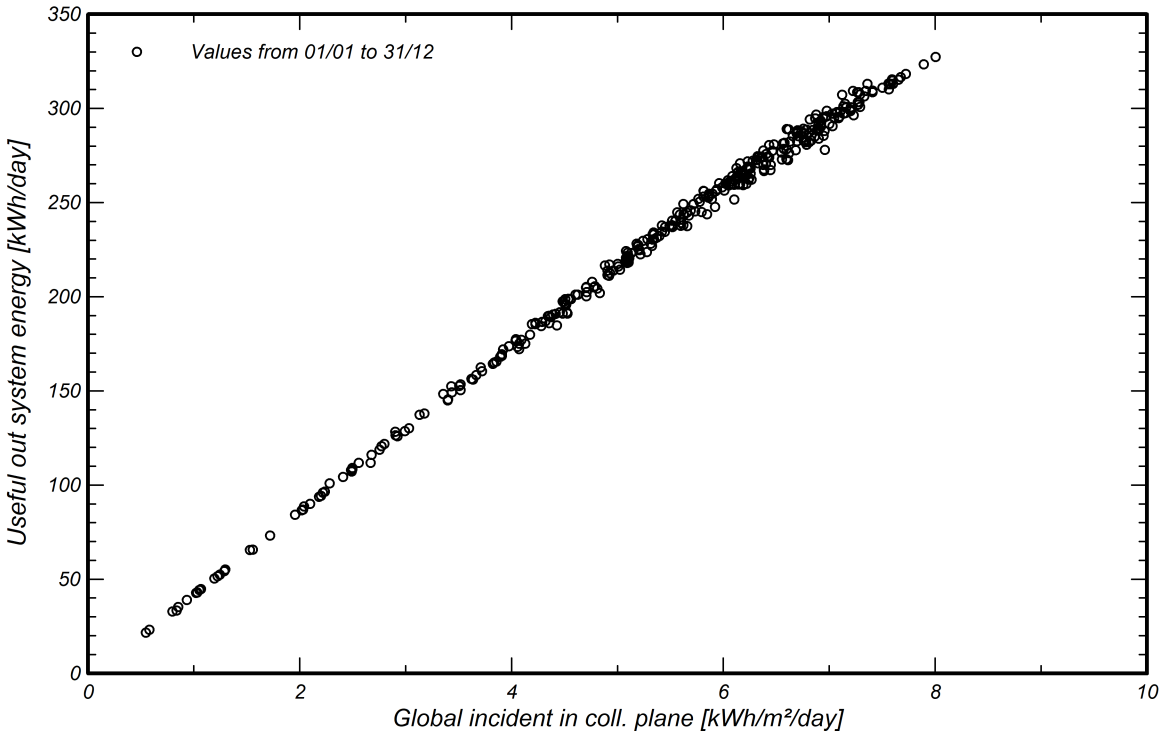
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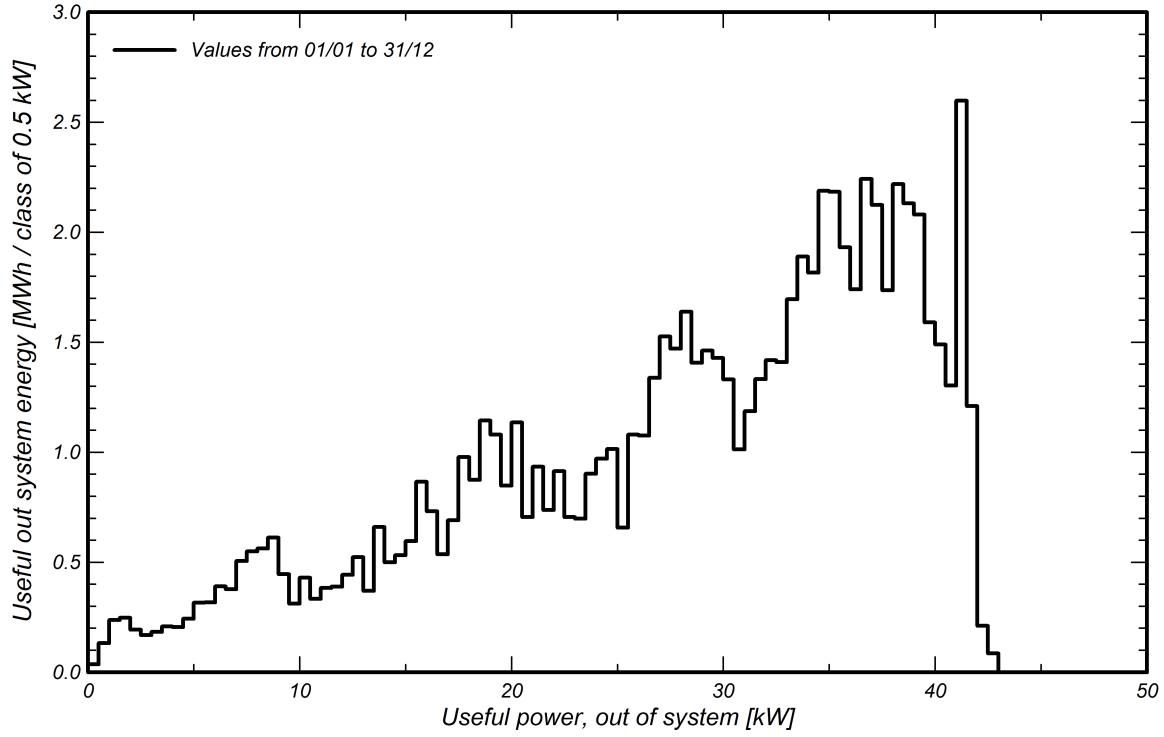
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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



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P50 - P90 evaluation

Weather data

Source	Meteonorm 9.0 dll, Sat=100%
Kind	Monthly averages
Synthetic - Multi-year average	
Year-to-year variability(Variance)	2.4 %

Specified Deviation

Mean energy shift	0.0 %

Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %

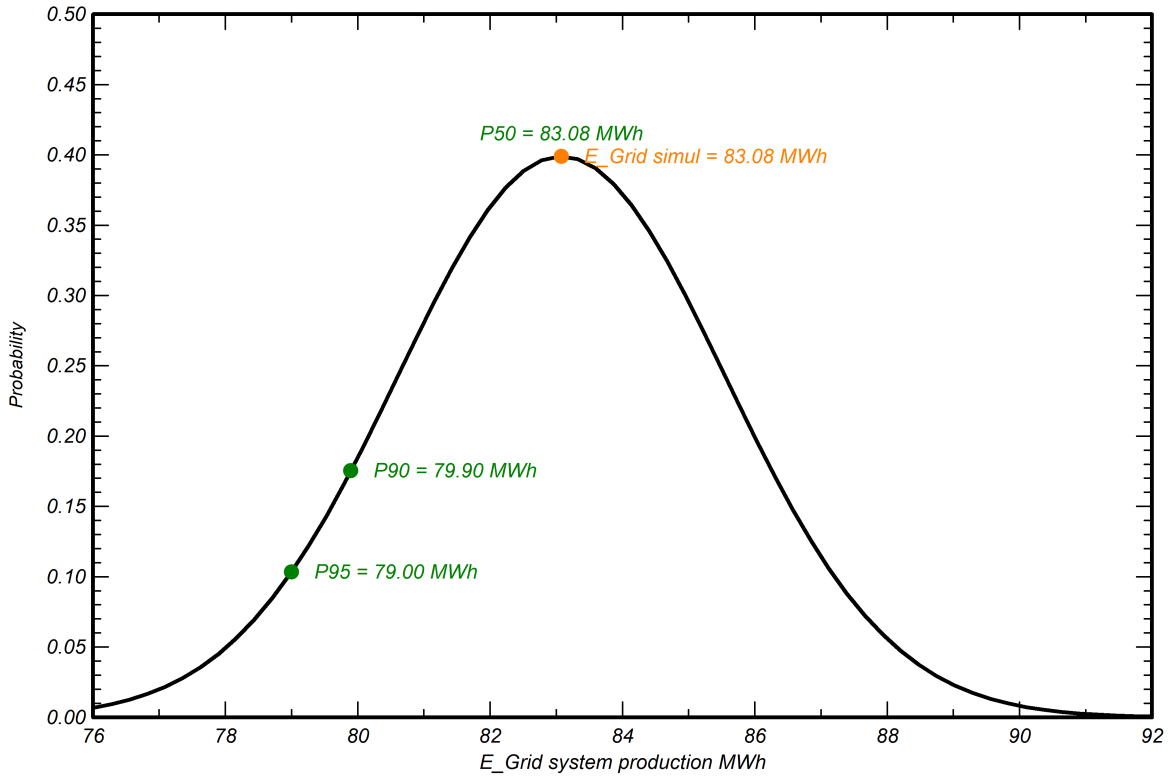
Global variability (weather data + system)

Variability (Quadratic sum)	3.0 %

Annual production probability

Variability	2.48 MWh
P50	83.08 MWh
P90	79.90 MWh
P95	79.00 MWh

Probability distribution



Cost of the system

Installation costs

Item	Quantity units	Cost EUR	Total EUR
		Total	0.00
		Depreciable asset	0.00

Operating costs

Item	Amount EUR	Payment schedule
Yearly average operating cost (OPEX)	0.00	

System summary

Total installation cost	0.00	EUR
Operating costs	0.00	EUR/year
Produced Energy	83.1	MWh/year
Cost of produced energy (LCOE)	0.0000	EUR/kWh

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CO_2 Emission Balance

Total: 1928.5 tCO_2

Generated emissions

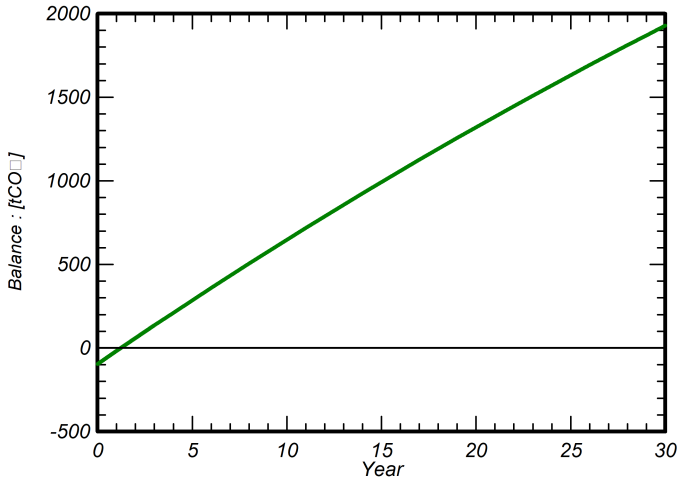
Total: 95.64 tCO_2

Source: Detailed calculation from table below

Replaced Emissions

Total: 2332.8 tCO_2
System production: 83.08 MWh/yr
Grid Lifecycle Emissions: 936 gCO_2/kWh
Source: IEA List
Country: India
Lifetime: 30 years
Annual degradation: 1.0 %

Saved CO_2 Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[$kgCO_2$]
Modules	1713 kgCO2/kWp	52.2 kWp	89404
Supports	6.24 kgCO2/kg	900 kg	5619
Inverters	619 kgCO2/-	1.00 -	619

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